STAT0023 Statistical Computing and Data Communication

Student handbook 2026/27 — Term 1 — 15 credits

Introduction

This module has two halves that are taught together rather than in sequence. The computing half covers the practical business of getting a statistical analysis to run, to run again a month later, and to run on somebody else's machine. The communication half covers what you then say about it. Students often arrive expecting the first half to be the hard one and leave having found the second harder.

Convenor: Dr R. Fairhurst, Department of Statistical Science. Teaching assistants are listed on Moodle. Questions about the material belong on the module forum rather than in email, so that the answer is visible to everyone.

Prerequisites

A first course in statistical inference and some previous exposure to programming, in any language. Students without the latter should expect the first three weeks to be demanding and should attend the optional drop-in sessions. Module registration closes on 9 October 2026 and cannot be extended.

Teaching arrangements

One lecture and one computer practical each week of Term 1, with no teaching in the reading week commencing 9 November 2026. Practicals are not assessed in themselves, but the in-course assessments draw directly on them, and students who skip practicals reliably find the assessments harder than they need to be.

All work for this module is done in R. The teaching version is R 4.4 with the tidyverse installed. If you work on your own machine, check your output against a cluster machine before you submit, because version differences have caused problems in previous years.

Assessment

The module is assessed by two pieces of in-course work and a take-home examination. The in-course work is designed to be done as the term progresses rather than in the week before each deadline.

In-course assessment 1 (30%)

A single analysis task with a supplied dataset, submitted as a reproducible R script and a short commentary. There is no word limit on the commentary, but anything beyond two pages suggests the analysis has not been thought through. The deadline is Term 1 Week 10. Note that this module numbers its weeks from the start of teaching, which is not the same as the numbering used on the central UCL timetable, so check the Moodle calendar if you are unsure which date that is.

In-course assessment 2: data communication report

You will take the analysis from the first assessment and rewrite it for a named non-specialist audience in 1,500 words, with at most three figures. The deadline falls two weeks after the reading week. The weighting of this component is confirmed on Moodle at the start of term once the cohort size is known; in recent years it has been between fifteen and twenty-five per cent.

Take-home examination (50%)

A three-hour take-home paper sat on Monday 11 January 2027 between 10:00 and 13:00, with a further thirty minutes allowed for submission. The paper is open book. You may use any written material and any software, but not another person. Past papers for the three preceding years are on Moodle, with solutions for the two oldest.

Submission

In-course work is submitted through the Moodle submission points, which close at 16:00 on the day of the deadline. Submit early: the submission point does not accept work after it closes, and a failed upload at 15:58 is not an extenuating circumstance. Keep a copy of the submission receipt.

Late work and extenuating circumstances

The standard departmental penalties apply. If illness or another serious disruption affects your work, submit an extenuating circumstances claim with evidence through Portico. Claims are considered by a panel, not by the convenor, and the convenor cannot grant an extension informally however reasonable the request.

Feedback and marks

Marks and written feedback for in-course assessment 1 are returned in the final week of term. Feedback on the second assessment is returned early in Term 2. Examination marks are released by Student Records with the rest of your results and are provisional until the board of examiners has met.